



Republic of the Philippines
Department of Agriculture
BUREAU OF FISHERIES AND AQUATIC RESOURCES
REGIONAL FISHERIES LABORATORY XII
J. Catolico St., Lagao, General Santos City

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Document Code: WI-01-CHE-03	MOISTURE CONTENT		

1. Purpose

To determine the percent moisture content in fish and fishery products.

2. Scope

This procedure is applicable to the determination of Percent Moisture Content in the meat of fish and fishery products.

3. Responsibility

The analyst performing the test should ensure that the procedure is followed accordingly.

4. Equipment/Laboratory Apparatus:

- 4.1 Convection oven (capable of heating up to 130°C)
- 4.2 Analytical Balance
- 4.3 Freezer
- 4.4 Desiccator (with desiccant)
- 4.5 Aluminum dish
- 4.6 Scissors
- 4.7 Crucible tongs
- 4.8 Beakers (150-ml)
- 4.9 Spatula
- 4.10 Forcep
- 4.11 Erlenmeyer Flask (250 ml)
- 4.12 Graduated cylinder (50 mL)

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4.13 Volumetric Flask

4.14 Watch Glass

4.15 Timer

6. Procedure

6.1 Pre-Analysis Procedure

1. Dish Taring
 - a. Take a clean dry dish.
 - b. Using a crucible tong, place the dish in an oven that is preheated to 100-105°C.
 - c. Heat the dish for one hour and fifteen minutes (1.15).
 - d. Cool the dish inside the dessicator.

6.2 Moisture Content Determination

1. Using a pair of tongs, weigh a tared dish.
2. Record the weight of the dish.
3. Tare the balance to turn the display into 0.0000 g.
4. Using a spatula, place 5grams of the sample in the aluminum dish.
5. Record the weight of the sample.
6. Put the dish inside an oven with the help of a tong.
7. Heat the sample for sixteen hours for raw octopus and one hour for dried samples at 100-105°C.
8. After heating, place the dish with the sample inside the dessicator.
9. Allow it to cool.

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10. Weigh the dish and the sample.
11. Record the weight of the dish and the sample.
12. Compute for the Percent Moisture Content.

7. Calculations

7.1 Percent Moisture Content

1. Compute for the Percent Moisture Content of the sample using Equation 1.

$$\text{Percent Moisture Content (\%MC)} = \frac{(x + y) - (z)}{y} \times 100 \quad (\text{Eq. 1})$$

where:

x = weight of the tared dish before heating.

y = weight of the sample before heating

z = weight of the dish with the sample after heating

7.2 Relative Percent Difference (RPD)

1. Compute for the RPD of duplicate trials using Equation 2.

$$\text{Relative Percent Difference (RPD)} = \frac{|a - b|}{\bar{x}} \times 100 \quad (\text{Eq. 2})$$

where:

$| |$ = absolute value

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$a = \%MC \text{ of trial } 1$

$b = \%MC \text{ of trial } 2$

$\bar{x} = \text{average } \%MC \text{ of trials } 1 \text{ and } 2$

$= (a+b)/2$

7.3 Relative Standard Deviation

1. Compute for the RSD of triplicate (or more) trials using Equation 3.

$$\text{Relative Standard Deviation (RSD)} = (s/x) * 100 \quad (\text{Eq. 3})$$

where:

$S = \text{standard deviation}$

$$= \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + (x_3 - \bar{x})^2 + \dots}{n - 1}}$$

$\bar{x} = \text{average value of the } \%MC \text{ of all the trials}$

$$= \frac{x_1 + x_2 + x_3 + x_4 + \dots}{n}$$

$x_1 = \%MC \text{ of trial } 1$

$x_2 = \%MC \text{ of trial } 2$

$x_3 = \%MC \text{ of trial } 3$

$n = \text{number of trials}$

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7.4 Sample Computation

1. Percent Moisture Content

Table 1. Raw Data for Percent Moisture Content Computation

Weight of Tared Dish (g)	Weight of Sample (g)	Weight of Tared Dish + Sample Before Heating (g)	Weight of Tared Dish + Sample After Heating (g)	Percent Moisture Content (%)
41.9837	1.2025	43.1862	42.2327	79.33

$$\begin{aligned}\text{Percent Moisture Content} &= \frac{(43.1862 - 42.2327)}{1.2025} \times 100 \\ &= 79.33 \%\end{aligned}$$

2. Relative Percent Difference

Table 2. Percent Moisture Content Data of Duplicate Analytical Trials

Trial No.	Percent Moisture Content (%)	Average Percent Moisture Content (%)	Relative Percent Difference (%)
1	79.33	79.48	0.38
2	79.63		

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$$\text{Relative Percent Difference} = \frac{(79.33 - 79.63)}{79.48} \times 100$$

$$= 0.38 \%$$

3. Relative Standard Deviation

Table 3. Percent Moisture Content Data of Five Analytical Trials

Trial No.	Percent Moisture Content (%)	Average Percent Moisture Content (%)	Standard Deviation	Relative Standard Deviation (%)
1	82.05	82.12	0.22	0.28
2	82.53			
3	82.05			
4	82.02			
5	81.97			

$$\text{Average Percent Moisture Content} = \frac{(82.05 + 82.53 + 82.05 + 82.02 + 81.97)}{5} \times 100$$

$$= 82.12 \%$$

$$\text{Standard Deviation} = \sqrt{\frac{[(82.05 - 82.12)^2 + (82.53 - 82.12)^2 + (82.02 - 82.12)^2 + (81.97 - 82.12)^2]}{(5-1)}}$$

$$= 0.22$$

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$$\text{Relative Standard Deviation} = \frac{(0.22)^8}{2.12} \times 100$$

$$= 0.28\%$$

10. Reference

AOAC Official Method 950.46 Moisture in Meat, Horwitz, et. al. (Editors), Official Methods of Analysis of AOAC International, 18th Ed., 2005.

Compiled Laboratory Procedures and Methods on CHEMICAL DETERMINATION AND SENSORY EVALUATION OF FISH AND FISHERY PRODUCTS, Chemical & Microbiological Section, Fisheries Post Harvest Technology Division, Bureau of Fisheries and Aquatic Resources, Quezon City, 1998.

11. FORMS

LF W01-CHE-02, Analyst Spreadsheet for Moisture Content Analysis

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12. DOCUMENT HISTORY

Revision No.	Issue Date	Original Entry	Revised as	Recall Date

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